AI and The Future of U.S Work

Exploring AI Occupational Exposure (AIOE), Median Wages, and Projected Employment Growth, 2024–2034

A Portfolio Project by Bella Boné

Introduction

In the year 2026 and beyond, as artificial intelligence continues to change the nature of work, understanding which occupations may be most exposed to AI can help workers, employers, and policymakers anticipate changes in the labor market. The following analysis examines AI Occupational Exposure (AIOE), wages, projected employment growth, and occupation categories to identify patterns associated with AIOE.

Research Questions

1) What is the relationship between an occupation's average earnings, projected employment growth, and AI exposure?
2) Do occupations with higher AI exposure share wage categories or occupational characteristics?
3) How can AI exposure, wages, and projected employment growth be used together to identify occupations that may be more vulnerable to automation?

Methodology

Occupation data from the U.S. Bureau of Labor Statistics and AIOE measures were cleaned, combined, and categorized using Excel. Exploratory data analysis was then conducted using Excel and Tableau to compare AI exposure, wages, education requirements, and projected employment growth across U.S. occupations.

Variables

- **AIOE (Score):** A standardized measure of an occupation's exposure to AI; higher values represent greater AI exposure.
- **SOC :** Occupational groupings within the federally classified Standard Occupational Classification (SOC) system.
- **Projected growth:** expected employment change
- **Wage Range:** Under $45K / $45K–$75K / $75K–$100K /$100K+

SOC Category Reference Key	
High AIOE/Cognitive	**Low AIOE/ Physical & Care Based**
23:Legal	29:Healthcare Prac
13:Business & Financ	31: Healthcare Support
15: Computer & Math	35: Food Prep & Service
25: Educational/Inst.	47: Construction
43: Office & Admin	51: Production

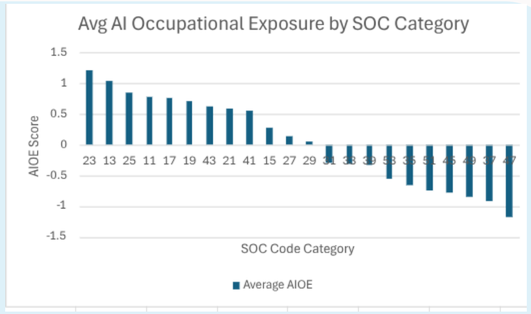


Figure1. Distribution of AI Occupational Exposure by SOC

Average AIOE varies substantially across occupational groups. Legal occupations (SOC 23) and Business & Financial Operations (SOC 13) show the highest average AI exposure, suggesting that cognitive, analytical, and information-processing work may experience greater AI exposure than more physical or hands-on occupations.

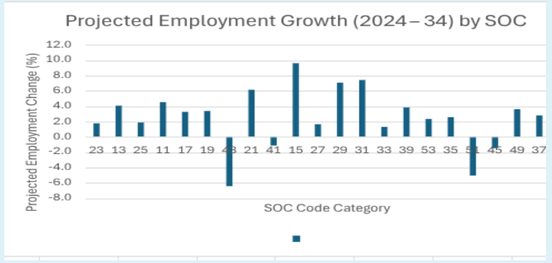


Figure 2. Projected Employment by Occupation (2024-2034)

Projected employment growth varies considerably across occupational groups. Computer & Mathematical occupations (SOC 15) show the strongest average projected growth at approximately 9.6%, while several other categories show flat or negative growth. Notably, high AI exposure can coexist with strong employment growth, suggesting that exposure alone does not indicate job decline.

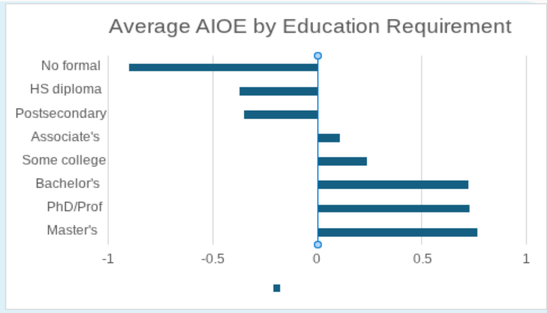


Figure 3. Average AIOE By education requirement

Average AIOE varies noticeably by typical education requirement. Occupations requiring a master's degree show the highest average AI exposure (~0.77), while occupations requiring some college but no degree (~−0.24) or an associate's degree (~−0.11) show lower exposure. Overall, higher educational requirements tend to correspond with greater AI exposure.

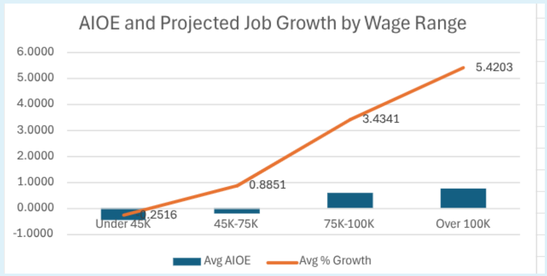


Figure 4. AIOE and Projected Job Growth by Wage Range

AI exposure and projected employment growth both increase across the higher wage ranges. Occupations earning over $100K show the highest average AIOE (~0.75) and strongest projected growth (~5.42%). This suggests that greater AI exposure does not necessarily indicate greater risk of job decline; in this dataset, the highest-wage occupations were both more AI-exposed and projected to grow more strongly.

Explore the Interactive Dashboard :
Scan here to view the full Tableau analysis ------>

Key Findings

01 — AI exposure varies substantially across occupations.

Legal and Business & Financial Operations occupations showed some of the highest average AIOE scores.

02 — Higher AI exposure does not necessarily signal job decline.

The relationship between AIOE and projected employment growth was relatively weak (**r ≈ −0.25**).

03 — Higher-paying and more highly educated occupations showed greater AI exposure.

Occupations earning over **$100K** had the highest average AIOE (**~0.75**), while master's-level occupations showed the highest exposure by education requirement (**~0.77**).

04 — AI exposure should be evaluated alongside other labor-market indicators.

Wages, education, occupational category, and projected growth provide important context for interpreting potential automation vulnerability.

Key Takeaway

AI exposure is not the same as job displacement. In this analysis, occupations with greater AI exposure were often higher-paying and did not consistently show weaker employment outlooks. Evaluating automation vulnerability therefore requires considering AI exposure alongside wages, projected growth, education, and occupational characteristics.

$100K+ occupations: AIOE ≈ 0.75 | Projected growth ≈ 5.42%
AIOE vs. projected growth: r ≈ −0.25

Conclusion

Overall, this analysis suggests that AI exposure should not be interpreted as a direct measure of job displacement risk. Higher AI exposure appeared most prominently in higher-paying, highly educated, and knowledge-based occupations, while several highly exposed occupational groups also maintained positive employment projections. Conversely, some lower-exposure occupations showed projected declines. Together, these findings suggest that career resilience depends on multiple factors—including wages, education, occupational characteristics, and demand—not AI exposure alone.

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